

Flexible Pouch – Product Specification Form

A complete explanation of every field in simple English for ERP development and user training.

1. Product Information Section

- PRODUCT ID – Auto-generated unique ID used for tracking job cards, BOM, costing, and production.
- PRODUCT NAME – Name of the pouch product, e.g., '1 Kg Atta Pouch', '1L Oil Pouch'.
- PRODUCT TYPE – Defines the packaging type, such as Laminated Pouch, Woven Sack, NonWoven Bag, Stand-up Pouch, Zipper Pouch.
- Laminate Type – Defines laminate quality such as Regular, High Barrier, Transparent.

2. Customer & Quantity Details

- CUSTOMER NAME – Auto-filled from customer master for traceability.
- QTY – Order quantity in KG, Pieces

3. Layer Structure Section

- NO OF LAYERS – Total layers (2, 3, or 4 layer structure).
- LAYER STRUCTURE – Combination such as PET 12 / METPET 12 / LLDPE 50.
- TOTAL MIC – Total micron thickness after adding all layers.

4. Product Size Details

- PRODUCT CLOSE SIZE – Final closed pouch size.
- WIDTH – Final pouch width.
- LENGTH – Final pouch length.
- GUSSET – Applicable for bottom/side gusset pouches.
- PRODUCT OPEN SIZE – Roll width before sealing.
- UPS – Units Per Sheet or per roll width for machine capacity.

5. Printing Details Section

- NO. OF COLOURS FRONT/BACK – Total print colors used on both sides.

- GSM / MIC – Printing layer thickness.
- DENSITY – Used for weight calculation (PET:1.38, BOPP:0.94, LLDPE:0.92).
- MATERIAL NAME – Film used for printing, such as PET/BOPP.
- SIZE W / SIZE L – Film width and length for printing.
- UPS – Units per roll width during printing.
- MACHINE – Printing machine number or line.

6. Ink and Solvent Details

- INK – Ink weight or coverage percentage.
- SOLVENT – Used in gravure printing, required for costing.

7. Lamination – 2nd Layer

- EXTRUSION / LAMINATION – Whether lamination is solvent-less, solvent-based, or extrusion type.
- GLUE – Adhesive quantity used.
- HARDENER – Required for solvent-based adhesive reaction.
- MATERIAL NAME – Second layer film (e.g., METPET, PE, CPP).
- MICRON – Thickness of second layer.

8. 3rd Layer Lamination

- Same parameters as 2nd layer: Micron, Material, Glue, Hardener.
- Usually LLDPE/LDPE for sealing layer.

9. Layer Weight (Auto Calculated) Calculated using formula:

$$\text{Layer Weight} = (\text{Micron} \times \text{Density} \times \text{Size W} \times \text{Size L}) / 1,000,000$$

10. Process Operation Section

- REWINDING / INSPECTION – If intermediate roll inspection is needed.

- SLITTING – Required to cut rolls to final width.
- POUCHING – Zipper, hole punch, gusseting, sealing operations.
- OPERATION COST – Additional operation or labour charges.

11. Pouch Weight

Auto-calculated total layer weight including ink and adhesive.

12. Costing Section

- Costing – Final cost per KG or per pouch.
- Conversion Cost – Machine, labour, electricity, overhead cost.
- Printing Cost – Based on colors used and ink consumption.
- Plus / Minus – Negotiation or adjustment amount.

13. Instructions (Clean Explanation)

1. Create product type masters such as Laminated Pouch, Woven Sack, etc.
2. Layer structure must auto-load based on laminate type.
3. Maintain density master for all film materials.
4. Maintain film material master with name, micron, GSM, rate.
5. Checkbox system required for operations like slitting, pouching.
6. Zipper calculation should include per meter or per piece cost.

ERP SPECIFICATION DOCUMENT

Flexible Pouch – BOM / Job Card / Costing Creation

1. Purpose of the Module

This module helps users create:

- Job Card (JDF)
- Bill of Material (BOM)
- Costing Sheet
- Process Routing
- Material Consumption Summary

It is designed for flexible pouch manufacturing.

2. Masters Required

A. Product Master

- Product ID, Name, Type
- Customer Name
- Layer Structure
- Total Micron
- Pouch Size
- Pouch Type

B. Material Master

- Films (PET, BOPP, LLDPE, METPET)
- Ink Shades
- Adhesive
- Cylinder Details

C. Process Master

- Printing
- Lamination
- Slitting
- Pouching
- Packing

3. Job Card (JDF) Fields

- Product ID
- Product Name
- Customer Name
- Quantity (KG/PCS/MTR)
- Price (optional)
- Number of Layers
- Total Micron
- Layer Structure

4. Layer Structure Importance

Each layer has:

- Material Type
- Micron
- GSM
- Wastage %

Material weight formula:

$\text{Material Weight} = (\text{Micron} \times \text{Density} \times \text{Width} \times \text{Length}) / 1,000,000$

Wastage is cumulative across processes.

5. BOM Auto Generation

The BOM will auto-add:

- PET Film
- METPET Film
- LLDPE
- CPP/BOPA

Adhesive Formula:

$\text{Adhesive (kg)} = (\text{Adhesive gsm} \times \text{Area}) / 1000$

Ink consumption based on coverage %.

6. Costing Sheet

A. Raw Material Cost

$\text{Material Cost} = \text{Material Weight} \times \text{Rate}$

B. Process Cost

Printing, Lamination, Slitting, Pouching

C. Cylinder Cost

Added or pro-rated

D. Total Cost/Kg

Total Cost = Material + Process + Wastage + Overheads

E. Margin

Selling Price – Total Cost

7. Process Routing

1. Printing – Color, Cylinder Repeat
2. Lamination – Layer Combination
3. Slitting – Final Roll Width
4. Pouching – Type and GSM

8. ERP Form Layout

Block 1: Product Details

Block 2: Structure Details

Block 3: Process Info

Block 4: BOM Auto Generated

Block 5: Costing
